SR-MPLS Training Lab

M6 Fast Reroute (TI-LFA)

Estimated time: 45 min

Objective

Give the SR-MPLS core sub-second protection against a single link or node failure using Topology-Independent Loop-Free Alternate (TI-LFA). You will confirm that IS-IS has already pre-computed a repair path for PE2's loopback, read the backup next-hop in both the routing table and the forwarding table, then fail PE1's plane-A primary link and watch traffic move onto the pre-programmed backup without waiting for IS-IS to re-converge. By the end you can prove protection is armed before a failure, not computed after one.

Before you start: M5 complete and the full SR-MPLS stack running: IS-IS Level-2 underlay, SR-MPLS transport with node-SIDs, the L3VPN overlay, Flex-Algo, and class-of-service all committed on every node so that 10.0.0.2 already resolves through an SR label-switched path.

Concept primer

TI-LFA is IS-IS fast reroute for a Segment Routing core. For every protected prefix, IS-IS runs a second SPF that removes the protected link (or node) from the topology and computes where traffic would flow once the network had fully re-converged. It then pre-installs that repair path in the forwarding table as a backup next-hop, expressed as an SR label stack. The key idea is that the backup is computed and programmed into hardware ahead of time, so when a link fails the line card simply switches to the already-present backup entry. Convergence is bounded by failure detection plus a hardware next-hop swap, which is why it is sub-second rather than the seconds a full IGP re-computation would take.

The repair path is built from SR segments, which is what makes it topology-independent. A classic LFA can only protect a prefix when a directly connected neighbor happens to have a loop-free path to the destination. TI-LFA instead encodes the post-convergence path as a stack of node and adjacency SIDs, steering the repaired packet along the exact path the network will use after re-convergence, even when that path is several hops away and no natural LFA exists. In this lab the label stack resolves toward PE2's node-SID 16002 over the surviving

plane, so the repaired traffic follows the same forwarding logic as the primary, just on the other path.

This behaviour is switched on by three lines that ship inside the ISIS-PARAMETERS group: 'interface <ge-*> level 2 post-convergence-lfa' arms TI-LFA on every ge- interface at once through the wildcard, 'backup-spf-options use-post-convergence-lfa' tells the backup SPF to prefer the post-convergence path, and 'backup-spf-options use-source-packet-routing' lets that backup be expressed as an SR label stack. Because they live in the group and are pulled in by apply-groups, one definition protects the whole fabric. They are immutable mandates and must never be negated. The other half of the design is physical: the P1-P2, P3-P4, and C1-C2 cross-links, all at metric 30, guarantee that when a primary plane-A link drops there is always a loop-free post-convergence path back onto plane B for the backup SPF to find. Without those cross-links TI-LFA could compute no valid repair for some prefixes, so the topology and the mandate work together.

Build steps

1. **Confirm the three TI-LFA mandates are present in the ISIS-PARAMETERS group.**

```
set groups ISIS-PARAMETERS protocols isis interface <ge-*> level 2 post-convergence-lfa
set groups ISIS-PARAMETERS protocols isis backup-spf-options use-post-convergence-lfa
set groups ISIS-PARAMETERS protocols isis backup-spf-options use-source-packet-routing
```

Note: These three lines are immutable mandates and are already present in the stage6-frr solution config on every SR node. The wildcard <ge-> arms post-convergence-lfa on all ge- interfaces at once, and the two backup-spf-options lines make the backup a post-convergence SR label stack. Do not negate or deactivate them. If they are missing, add them and 'commit'.*

2. **Verify a backup next-hop for PE2 is already pre-computed before any failure.**

```
lab@PE1> show isis route 10.0.0.2 extensive    ## look for TILFA computed MPLS paths Valid:1
lab@PE1> show route 10.0.0.2 detail            ## look for a Backup next-hop under the primary
```

Note: Protection must be armed with the network healthy. Both plane-A (ge-0/0/1 toward P1) and plane-B (ge-0/0/2 toward P2) primaries are present, and the TI-LFA backup is pre-installed. If there is no Backup next-hop now, the demonstration below will drop traffic.

3. **Fail the plane-A primary link on PE1 and observe the backup take over.**

```
lab@PE1> configure
lab@PE1# set interfaces ge-0/0/1 disable
lab@PE1# commit and-quit    ## or, in GNS3, right-click the PE1-P1 link and Stop it
lab@PE1> show route 10.0.0.2 detail    ## traffic now on the former backup path via ge-0/0/2
```

Note: Disabling ge-0/0/1 (or stopping the PE1-P1 link in GNS3) fails the plane-A primary. Because the backup was pre-programmed, forwarding moves onto plane B in sub-second time rather than after a full IGP re-convergence. Watch a running 'ping 10.0.0.2 source 10.0.0.1 rapid' lose at most a packet or two.

4. **Restore the primary link and confirm protection re-arms.**

```
lab@PE1> configure
lab@PE1# delete interfaces ge-0/0/1 disable
lab@PE1# commit and-quit    ## or Start the PE1-P1 link in GNS3
lab@PE1> show route 10.0.0.2 detail    ## primary via ge-0/0/1 with the Backup next-hop restored
```

Note: Always return the lab to full protection. Each node's stage6-frr config carries the same three group lines, so a fabric-wide reset is 'python push_lab_configs.py stage6-frr'. PE1 and P1 are shown as representatives; the mandate is identical on every SR node.

Verification

show isis route 10.0.0.2 extensive

```
10.0.0.2/32  L2  metric 40
 ge-0/0/1.0  IPV4 P1
 ge-0/0/2.0  IPV4 P2
 TILFA computed MPLS paths, Count:1, Valid:1
    PathId:0, SID count:1
```

What to look for: 'TILFA computed MPLS paths' with 'Valid:1' is the proof that IS-IS has pre-computed a usable post-convergence backup toward PE2's node-SID 16002. 'Valid:0' means the use-post-convergence-lfa mandate is missing or deactivated and no backup exists.

show route 10.0.0.2 detail

```
10.0.0.2/32 (1 entry, 1 announced)
      *IS-IS Preference: 18
       Next hop: 10.1.1.2 via ge-0/0/1.0, selected   ## primary, plane A via P1
       Next hop: 10.1.1.18 via ge-0/0/2.0          ## Backup next-hop, plane B via P2
         Label operation: Push 16002
       Backup next hop found by TI-LFA
```

What to look for: The selected primary points out ge-0/0/1 toward P1 on plane A, and a second Backup next-hop out ge-0/0/2 toward P2 on plane B is listed beneath it. A primary with no Backup next-hop means protection is not armed.

show route forwarding-table destination 10.0.0.2

```
Destination       Type RtRef Next hop           Type Index
10.0.0.2/32       user     0                    ulst
                           10.1.1.2  ge-0/0/1.0  ucst   (primary)
                           10.1.1.18 ge-0/0/2.0  ucst   (backup)
```

What to look for: The backup is programmed into the FIB itself, not just the control plane. Seeing both next-hops in the forwarding-table is what makes the failover a hardware swap and therefore sub-second. If only the primary appears here, the line card has nothing to fail over to.

show route 10.0.0.2 detail ## run again after 'set interfaces ge-0/0/1 disable ; commit'

```
10.0.0.2/32 (1 entry, 1 announced)
      *IS-IS Preference: 18
       Next hop: 10.1.1.18 via ge-0/0/2.0, selected   ## now primary, plane B via P2
        Label operation: Push 16002
```

What to look for: After the plane-A link is disabled, the former backup out ge-0/0/2 is now the selected primary and traffic to PE2 continues on plane B. The switchover happened before IS-IS ran a fresh SPF, which is the whole point of pre-computing the repair path. Re-enable ge-0/0/1 when done.

[screenshot 007: M6 verification: TI-LFA precomputed backup next-hop for PE2]

Break / Fix exercise

Scenario: The lab looks healthy and 10.0.0.2 is reachable, but during a maintenance test the team disabled PE1's plane-A link and watched a steady ping to PE2 drop packets for several seconds before recovering, instead of the sub-second blip the design promises. Every adjacency is Up and the SR label-switched path to PE2 works, so nothing looks wrong until a link actually fails.

Your task: Explain why traffic was unprotected and restore sub-second fast reroute for PE2's loopback across the fabric. Confirm your fix by proving a backup next-hop is pre-computed again before the next failure test.

Hint: With the network healthy, check 'show isis route 10.0.0.2 extensive' for 'TILFA computed MPLS paths' and its Valid count, and 'show route 10.0.0.2 detail' for a

Backup next-hop. A Valid:0 result and a missing backup, with the underlying stanza still visible in the config, is the signature of a group-level mandate that has been negated rather than removed. This is a mandate-negation fault; the correct fix restores the immutable TI-LFA mandate in the ISIS-PARAMETERS group and re-arms protection on every IS-IS interface at once. Compare 'show configuration groups ISIS-PARAMETERS | display inheritance' against the stage6-frr solution to spot which mandate is inactive.

The instructor holds the fault-injection and fix commands in the answer key. Diagnose it yourself first.

Brief it back

Explain each of these aloud before moving on. If you cannot, re-read the primer and re-run the verification commands.

- Explain how TI-LFA pre-computes a loop-free backup ahead of any failure and how it encodes that repair path as an SR label stack toward the destination's node-SID.
- Why does deactivating one line in the ISIS-PARAMETERS group remove protection on every IS-IS interface in the fabric at the same time, and why is that so easy to miss in a group-heavy config?
- Describe the role of the metric-30 cross-links (P1-P2, P3-P4, C1-C2) in guaranteeing that the backup SPF can always find a loop-free post-convergence path for the repair.